

M. Awais Aslam

AI/ML Engineer / Data Scientist / Python Developer

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Summary

AI/ML Engineer with 1.5 year of experience in Gen-AI, and Python development. Skilled in designing and deploying AI systems, with strong expertise in Python development, pipeline engineering, and scalable solution delivery.

Experience

Associate Python/AI Developer – Insightlix Pvt Ltd

Jan 2026 – May 2026

- Brain Tumor Detection System (Django + DL): Built a deep learning pipeline for MRI analysis and tumor detection (segmentation of edema and lesions) using SynthSeg and nnUNet.
- Real Estate AI Agent (Django): Developed a multi-tenant conversational AI platform with customizable workflows, dynamic forms, and plan-based user configurations.
- Showtime Validation & Classification (FastAPI): Built APIs for extracting and validating movie showtimes; automated Excel validation and amenity classification.
- Enterprise ERP Platform (Django): Built an accounting system with dynamic billing workflows, secure payments, and client management.
- Configured PageIndex for document indexing and retrieval workflows.
- Deployed and customized OpenClaw AI Assistant solutions for client demos.
- Integrated Stripe for secure payments and subscription management.

Associate Data Scientist – Programmers Force

Sept 2025 – Dec 2025

- Developed computer vision models for document verification (text extraction & forgery detection) using:
 - OCR, ELA, and noise analysis with ResNet, EfficientNet, and ViT.
 - Designed ensemble pipelines with parallel inference for extraction and forgery detection.
- Developing models for Facial Verification & Liveness detection:
 - Implemented liveness detection mechanisms to prevent replay and spoofing attacks.
 - Fine-tuned models on large-scale datasets for improved real-time accuracy and performance.
- Contributed to enhancing Shufti's identity verification system through development and testing of CV models.

Associate AI Developer – Octaloo Technologies

June 2025 – Aug 2025

- Managed backend infrastructure for a Restaurant Order Management System using Django and SQLite, optimizing performance and security with JWT and RBAC.
- Developed a communication platform (Slack-like) using Flask with WebSocket-based messaging and notifications.
 - Designed scalable multi-tenant APIs for workspace, channel, super admin, admin, and user management.
 - Integrated chat analyzer (sentiment/trends) and LLM-based next-word prediction.
- Built RESTful APIs with Flask and FastAPI, and Django integrating AI/ML models into production workflows.

Python/AI Intern – Octaloo Technologies

March 2025 - May 2025

- Built AI chatbots using FastAPI, MongoDB, and Ollama with LoRA, QLoRA, quantization, and RLHF.
- Developed LLM workflows using LangChain tools (LangGraph, LangServe, LangFlow, LangSmith) for prompt design and agent orchestration.
- Created AI agents, including Basic Agent, Document RAG, Financial Analyst, and Video Summarizer.
- Contributed to an AI-powered pay-per-minute call platform:
 - Integrated voice agents, Stripe payments, RBAC, and Twilio call logs via FastAPI.
 - Developed custom and knowledge-based agent workflows with OTP authentication.

Machine Learning Intern – PITB

Dec2024 - Jan2025

- Developed computer vision and deep learning solutions for real-world applications.
- Gained hands-on experience with OpenCV, YOLO-based object detection, and OCR techniques.
- Optimized detection and recognition models for improved accuracy and performance.
- Built an automatic number plate recognition (ANPR) system.
- Utilized OpenCV for image preprocessing and real-time video processing.

Data Science Intern – Ecodecamp

Sep 2024

- Performed EDA, predictive modeling, and customer segmentation to support data-driven decisions.
- Enhanced skills in Python, SQL, and machine learning through hands-on projects.
- Applied data science techniques to solve real-world problems.
- Generated actionable insights using data visualizations and analytical reports.

Skills

- **Languages:** Python | JavaScript | C++
- **AI/ML:** Machine Learning | Deep Learning | Generative AI | NLP | Computer Vision | Fine-Tuning | RAG | MCP
- **Frameworks:** TensorFlow | PyTorch | Scikit-learn | LangChain | LangGraph | LangSmith | LangFlow | LangServe | LlamaIndex | Agno (Phidata) | FastAPI | Flask | Django | Streamlit
- **Databases:** MongoDB | MySQL | PostgreSQL | Pinecone | ChromaDB | FAISS
- **Visualization:** Power BI | Matplotlib | Seaborn | Plotly
- **Tools & Libraries:** Git | GitHub | Docker | Selenium | Playwright | BeautifulSoup | Celery | Ngrok | PageIndex

Projects

Communication Tool-like Slack (Flask):

- Built a real-time communication platform using Flask, WebSockets, REST APIs, JWT, and SQLite.
- Integrated AI models for next-word prediction and chat analysis.

AI-Powered White Label Pay-Per-Minute Call Platform (FastAPI):

- Contributed to an AI-powered pay-per-minute calling platform with FastAPI, Stripe, Twilio, and RBAC.
- Developed AI voice agents with RAG-based knowledge base workflows and OTP authentication.

Brain Tumor Detection Model (Django):

- Built a deep learning pipeline for MRI-based brain tumor detection and segmentation.
- Used SynthSeg & nnUNetV2 for segmentation and clinical insight generation.

Automatic Number Plate Recognition System (Flask):

- Contributed to an ANPR system using Flask and Hugging Face models for vehicle and number plate detection.
- Implemented vehicle attribute recognition (plate, color, category, model) with SQLite storage.

AI News System (Django & React):

- Built an AI news platform for web scraping, article storage, and LLM-based summarization.
- Integrated gTTS for audio news generation with a Django & React dashboard.

Restaurant Order Management System (Django):

- Developed a Django-based Restaurant Order Management System with QR code menu access.
- Implemented customer order placement and order routing to the kitchen with order tracking.

MCP-MultiTool AI Agent (LangChain):

- Developed an MCP-based multi-tool AI agent using Google API, supporting dynamic task handling.
- Integrated chatbot, mathematical computation, and weather assistant as callable tools.

Document Reader RAG-Agent (LangChain): Developed a LangChain-based RAG agent using the Google Gemini API, enabling users to upload documents and query them conversationally.

Video Summarizer RAG-Agent (Phidata - Agno): Built a LangChain-based RAG agent with the Google Gemini API to analyze uploaded videos and answer user queries.

OTHER PROJECTS: OpenClaw AI Assistan | Real Estate AI Agent | PageIndex RAG Chatbot | Medical Chatbot | Breast Cancer Predictor | Automation Web-Scrapping | Next Word Predictor | SMS Spam Classifier | Image Classifier | Posture Detection | Face-Mask Detection | Chat Analyzer | Match Win Probability System | Document Verification System | Facail Verification System | Facial liveness Detection | Enterprise Resource Planning (ERP) System

Achievements

- Developed and deployed scalable AI-powered backend systems using FastAPI, Flask, and Django.
- Built and integrated RAG-based AI agents with LLM workflows for automation and intelligent assistance.
- Fine-tuned transformer models using LoRA/QLoRA for domain-specific NLP tasks.
- Implemented real-time communication and authentication systems using WebSockets and JWT.
- Integrated machine learning and computer vision models into production-ready applications.
- Automated data extraction, validation, and payment workflows to improve operational efficiency.

Certifications

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| • Python for AI Coursera Jan 2023 | • Data Science Internship Ecodecamp Sept 2024 |
| • Data Science CampusX Feb 2023 | • ML Engineer Internship PITB Dec 2024 |
| • Intro to ML/DL CampusX Jan 2024 | • AI Developer Internship Octaloo Tech March 2025 |

Education

F.S.C Pre-Engineering
Govt. Degree College
2018-2020

Bachelor's Degree in Computer Science
NCBA&E
Feb 2021 - Jan 2025